MOTIVATION · THE PROBLEM

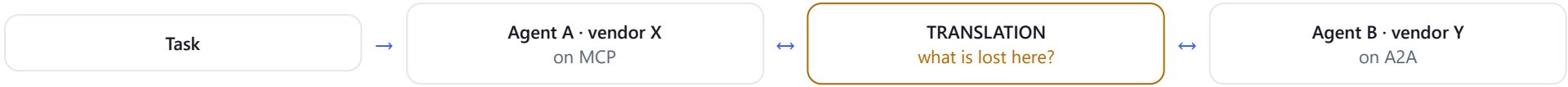
Agents are multiplying across vendors. Nothing measures whether they cooperate.

WHAT CHANGED IN 2024 TO 2025

- **MCP** (Anthropic, 2024) standardized how an agent reaches tools and context.
- **A2A** (Google, 2025, now Linux Foundation) standardized how agents discover and hand off tasks to each other.
- Production stacks now mix agents from OpenAI, Anthropic, Google, Meta, and open models.

WHAT IS STILL MISSING

- Protocols standardize **transport**, not the **meaning** that survives a handoff.
- Vendor specs claim interoperability but ship **no neutral measurement**.
- When a task crosses a protocol or vendor boundary, we cannot yet say **how much is lost**.



The red box is the unmeasured seam. Our benchmark turns it into numbers: task success, context preserved, capability preserved, latency, overhead, error recovery.

RESEARCH QUESTIONS · CONTRIBUTIONS

What we ask, and what we deliver

RESEARCH QUESTIONS

- **RQ1.** When agents from different vendors interoperate across MCP and A2A, how often do end-to-end tasks still succeed?
- **RQ2.** How much task-relevant context and how many advertised capabilities survive a cross-protocol handoff?
- **RQ3.** What are the dominant failure modes, and how well do systems recover from them?

CONTRIBUTIONS (THE NOVELTY)

- **C1.** The first **neutral, cross-vendor** interoperability benchmark centered on the MCP and A2A boundary.
- **C2.** A metric for **context and capability preservation** through protocol translation, not just end success.
- **C3.** A reproducible, contamination-resistant harness with a public task set and baselines.

Novelty statement: existing work either compares protocols on generic engineering axes (ProtocolBench) or documents that meaning degrades across LLM handoffs (MAST, boundary-metadata collapse, semantic-view), but no benchmark isolates the cross-vendor boundary itself. We contribute the first benchmark that pairs different-vendor agents across MCP and A2A and quantifies both task success and context preservation through the translation layer, reframing the classic FIPA and KQML meaning-preservation problem for the LLM era.

BACKGROUND · PROTOCOLS AND THEIR HISTORY

The interoperability stack, old and new

TODAY'S AGENT PROTOCOLS

- **MCP** (Anthropic): agent to tool and context. Lightweight, request-response.
- **A2A** (Google, Linux Foundation): agent to agent discovery and task handoff via Agent Cards. Stateful, richer, higher overhead.
- **ACP, ANP**: further proposals covered in the protocol survey (2505.02279).

HISTORICAL GROUNDING (RELATED WORK)

- **FIPA ACL** (1997 to 2002) and **KQML**: speech-act performatives for agent messaging.
- They assumed shared ontologies and symbolic logic, so meaning was well-defined by construction.
- LLM agents are probabilistic and opaque, so the old meaning-preservation guarantee is gone. That is precisely the gap we measure.

1997 · FIPA / KQML
symbolic, shared ontology

→

2023 · multi-agent LLM frameworks
homogeneous, single-vendor

→

2024 to 2025 · MCP, A2A
cross-vendor transport, no metric

Inference: the field solved *transport* interoperability (agents can connect) but not *semantic* interoperability (the right meaning arrives). The semantic-view paper (2604.02369, 2026, verify) makes this argument explicitly.

LITERATURE 1 OF 3 · INTEROPERABILITY PROTOCOLS AND THEIR EVALUATION

What has been studied about the protocols themselves

SURVEYS AND COMPARISONS

Survey of agent interop protocols MCP, ACP, A2A, ANP (2505.02279). Canonical taxonomy, but purely qualitative: no measurement of what breaks.

MCP vs A2A comparative study (2607.23884, 2026 verify). Single-team implementation; does not test cross-vendor agents or quantify loss.

Semantic view of agent protocols (2604.02369, 2026 verify). Names the semantic gap conceptually; no benchmark to measure it.

GOVERNANCE AND SECURITY

Governance gaps (2606.31498, 2026 verify) · **Security threat modeling** (2602.11327, 2026 verify) · **MCP server security study** (2506.13538). Policy and attack-surface lenses; nothing on legitimate cross-vendor task outcomes.

CLOSEST PRIOR ART

ProtocolBench "Which multi-agent protocol to choose?" (2510.17149). Benchmarks protocols on success, latency, overhead, robustness. Our differentiator: it does not isolate context and capability preservation across a cross-vendor MCP to A2A boundary.

Takeaway: the protocols are surveyed, secured, and compared, but interoperability is still treated as an engineering property, not a measured outcome. The one true benchmark (ProtocolBench) compares protocols, not vendors crossing them.

LITERATURE 2 OF 3 · MULTI-AGENT SYSTEMS AND HANDOFF LOSS

Systems that could be tested, and evidence that meaning leaks

MULTI-AGENT FRAMEWORKS (SYSTEMS UNDER TEST)

MetaGPT (2308.00352) · **AutoGen** (2308.08155) · **CAMEL** (2303.17760) · **ChatDev** (2307.07924). All powerful, but agents are homogeneous and single-framework. Cross-vendor interop is untested.

MultiAgentBench (2503.01935). Measures collaboration and competition, not protocol interop or vendor heterogeneity.

BFCL function-calling leaderboard (PMLR v267). Cross-vendor tool-call fidelity, but single-agent, not multi-hop across a handoff.

EVIDENCE THAT CONTEXT IS LOST ACROSS HANDOFFS

Why do multi-agent LLM systems fail? MAST taxonomy (2503.13657). 1,600+ traces; about 37% of failures are inter-agent misalignment. Diagnostic, not a metric suite.

Lost in the Middle (2307.03172). Models drop mid-context information. Motivates cumulative loss across handoffs.

Boundary metadata collapse in handoffs (2608.29028, 2026 verify). Summary handoffs keep facts but drop usage constraints.

Mixed-vendor multi-agent (clinical) (2603.04421, 2026 verify). Rare cross-vendor study: mixed teams help, but no interop metric.

Takeaway: multiple independent papers show meaning and constraints degrade across agent handoffs, yet none provide a standardized preservation metric across a protocol boundary. The evidence for the problem exists; the measurement does not.

LITERATURE 3 OF 3 · BENCHMARK METHODOLOGY AND RIGOR

What makes a benchmark credible and publishable

DESIGN TEMPLATES

HELM (2211.09110): scenario times metric taxonomy, multi-metric per scenario. Model to emulate: score several axes, not one number.

SWE-bench (2310.06770) · **GAIA** (2311.12983) · **WebArena** (2307.13854) · **tau-bench** (2406.12045). Execution or state grading and reset-able envs give objective, reproducible scores.

ToolBench (2307.16789) · **API-Bank** (2304.08244). Decompose a capability into sub-skills and score each.

VALIDITY AND PITFALLS

AI Agents That Matter (2407.01502): report cost, use holdout, standardize harness. **ABC checklist** (2507.02825, 2025 verify): task-validity flaws move scores by up to ~100%. **Emergent mirage** (2304.15004) and **Everything benchmark** (2111.15366): metric choice and construct validity. **LLM-as-judge** (2306.05685): judges are biased.

CONTAMINATION AND STATISTICS

LiveBench (2406.19314) · **LiveCodeBench** (2403.07974): timestamped, refreshed tasks resist leakage. **pass@k** (2107.03374) and **error bars for evals** (2411.00640): report reliability and confidence intervals.

Takeaway: a benchmark is credible when it has objective grading, construct validity, reproducibility, contamination resistance, and honest statistics. These are the pillars our design must satisfy (slide 9).

GAP ANALYSIS · THE WHITESPACE

Coverage matrix: no one occupies the cross-vendor row

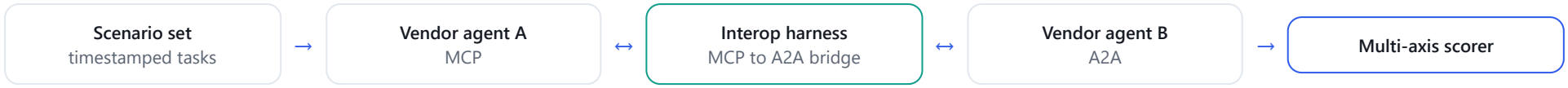
Work cluster	Task success	Cost / latency	Context preserv.	Capability preserv.	Cross-vendor	Contam. resistant
Protocol surveys (2505.02279)	no	no	no	no	part	no
ProtocolBench (2510.17149)	yes	yes	no	no	no	no
MultiAgentBench (2503.01935)	yes	no	part	no	no	no
BFCL (function calling)	yes	no	no	part	part	part
Handoff-loss studies (MAST, 2608.29028)	no	no	part	no	no	no
This proposal	yes	yes	yes	yes	yes	yes

THE OPEN GAPS

- **G1.** No neutral cross-vendor interoperability benchmark exists.
- **G2.** Context and capability preservation are discussed but never measured across a protocol boundary.
- **G3.** Framework benchmarks assume homogeneous, single-vendor agents.
- **G4.** Tool-call fidelity is measured single-agent only, never surviving a cross-protocol hop.

PROPOSED FRAMEWORK · DESIGN AND METRICS

The benchmark, and why each choice is defensible



SIX METRICS

- **Task success** (execution or state graded, not judged).
- **Context preservation**: key-fact recall of the originating context after the hop.
- **Capability preservation**: advertised tools or Agent Card still invocable and correctly typed.
- **Latency** and **communication overhead** (messages, bytes) per task.
- **Error recovery** under injected faults (dropped context, capability mismatch).

FIVE DESIGN PRINCIPLES, EACH GROUNDED

- **Objective grading**: execution and AST or structural matching (SWE-bench, BFCL).
- **Multi-axis scoring**, not one number (HELM).
- **Contamination resistance**: timestamped, refreshable tasks (LiveBench, LiveCodeBench).
- **Reproducibility and cost** reporting, standard harness (AI Agents That Matter, ABC checklist).
- **Reliability**: pass@k, pass^k, confidence intervals (HumanEval, tau-bench, error-bars paper).

PLAN · FEASIBILITY · VENUES · HONESTY

Why this finishes, where it goes, and what we cannot yet claim

PHASED PLAN

- **P1.** Define 2 to 3 interop scenarios and the decision or task contract.
- **P2.** Build the MCP to A2A harness and the six metrics.
- **P3.** Run 3 to 4 vendor agents, collect baselines.
- **P4.** Analysis, error bars, write-up.

FEASIBILITY (SCOPED FOR ONE YEAR)

- Measurement first: no new protocol or negotiation engine to invent.
- Start with 2 or 3 scenarios, expand only if time allows.
- Reuse public agents and reset-able environments.

TARGET VENUES (NEAR TO ASPIRATIONAL)

- Workshop at NeurIPS, ICLR, or ACL (agents or evaluation themed).
- Aspirational: NeurIPS Datasets and Benchmarks track.
- Resource tracks at ACL, EMNLP, NAACL.

Stated honestly, for your review: several 2026 papers cited (semantic view, governance, security, boundary-metadata, mixed-vendor clinical, and the MCP vs A2A comparison) were confirmed by search snippets but their exact arXiv IDs, authors, and venues should be verified on the abstract pages before the final reference list. Landmark works (HELM, SWE-bench, GAIA, MMLU, FIPA, MetaGPT, AutoGen, CAMEL, ChatDev, Lost-in-the-Middle, BFCL, LiveBench, pass@k) are well established and safe to cite. One open feasibility risk: we must confirm that enough public vendor agents expose usable MCP and A2A endpoints for a clean test set. That check is step P1.